A-A-1899
November, 24, 1986

COMMERCIAL ITEM DESCRIPTION

CUSHIONING MATERIAL, PACKAGING (CELLULOSIC, WATER RESISTANT)

The General Services Administration has authorized the use of this commercial item description in preference to Type II, Classes A (compressed) and B (uncompressed) of PPP-C-843D.

This description covers (Class A) low tensile strength filler material and (Class B) high tensile strength wrapping material used for cushioning fragile containers and for protecting containers from liquids broken in transit.

Salient characteristics.

The cushioning material shall be composed of cellulosic material that will provide a product complying with the requirements of this commercial item description.

The cushioning material shall be plain, uncompressed or longitudinally compressed, roll or multifold units, and perforated style. For plain (uncompressed) material, perforations are a linear series of holes in the material to permit tearing off of a premeasured length of material.

Absorbent capacity.

The cushioning material shall absorb not more than 3 times its weight of water when tested as follows:

Weigh to 0.1 grams (i.e., to nearest 0.01 and round off to tenths) a 4 ± 1/8 inch square piece of material having a thickness of not less than 1/2 inch. This is the dry weight (thinner material shall be stacked to at least 1/2 inch thick and stapled with preweighed staples at the four corners). Place the material on the surface of water which is maintained at 21-24 degrees C (70-75 degrees F). The water bath shall be sufficiently large to permit the material to submerge. After 30 seconds, remove the material with a piece of 1/2 inch mesh screen which is 6 inches square. Drain in a horizontal position for 1 minute and weigh the wet sample as described above. This is the wet weight. Calculate the absorbing capacity as follows:

$$\text{Absorbent capacity (no. of times its dry wt.)} = \frac{\text{wet. wt. - dry wt. - (staple wt. if applicable)}}{\text{dry wt. - (staple wt. if applicable)}}$$

Compression set.

The material shall have a compression set of not more than 40 percent when tested as follows:

Place a stack of 4-inch square (± 1/8-inch) samples of material having a minimum of 1-inch measured thickness between parallel, rigid, flat plates. Compress to 50 percent of the measured thickness for 2 hours. Three minutes after the pressure has been removed, measure the final thickness. Calculate as follows:

$$\text{Compression set (in percent)} = \left(\frac{\text{original thickness - final thickness}}{\text{original thickness}}\right) \times 100$$

FSC 8135

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:17Z
Check the source to verify that this is the current version before use.

The longitudinally compressed material shall have the capability of expansion (decompression) to not less than 4.8 times its length without rupture of the plies. The longitudinally compressed material shall be measured for length in the compressed state.

For the longitudinally compressed material, expansion shall be determined by placing a sample (roll or multifold unit) on a level surface, unroll or unfold and measure the entire length in the compressed state. Cut a 4 by 14-inch specimen and suspend it from a clamp having flat faces 2 by 4-inches or greater. At a distance of 10 inches (accurately measured) attach another clamp similar to one above and either add sufficient weight or pull down by hand to its maximum distance without rupture of plies. Measure accurately the expanded length of the specimen between the clamps. Calculate the expansion as follows:

$$\text{Expansion} = \frac{\text{expanded length}}{\text{compressed length}}$$

The width of the roll shall be as specified with a tolerance of plus or minus 3 percent or 1/4-inch, whichever is larger. The width of the unrolled rolls shall be the average of four measurements. For longitudinally compressed material, perforations are cut out areas leaving one inch linkages along the width to permit the tearing off of a premeasured length of material.

Measured thickness

The measured thickness of the material shall be not less than 85 percent of the specified thickness.

Place a stack of 4-inch square (± 1/8-inch) samples of material which is at least 1 inch high on a level surface. If sample is 1 inch thick or greater, one sample is sufficient. Place a 6-inch square flat weight plate which weighs 0.4 ± 0.005 pounds on top of this stack. After 1 minute, measure the vertical distance (thickness) between the level surface and the weight plate at each of the four corners and record the average (final thickness). The average of the four measurements shall be the measured thickness. (Note: If more than one piece of material is used, the measured thickness shall be the final thickness divided by the number of pieces). The percent of specified thickness (as specified in the contract) shall be calculated as follows:

$$\text{Percent of specified thickness} = \frac{\text{(measured thickness)}}{\text{(specified thickness)}} \times 100$$

Strain limit. The strain limits of the material shall be not less than 50.0 nor more than 75.0 percent when tested as follows:

Place a sample(s) of material to form a stack of 4-inch square (± 1/8-inch) material at least 1-inch high. If sample is 1-inch or greater, one sample is sufficient. Measure the height (measured thickness) of this stack. (Note: For the purpose of this test, if each sample is less than 1-inch in measured thickness, the measured thickness of the stack and not the individual samples is used.) Place a 3 p.s.i. load on this stack for 15 seconds. Measure the loaded height at the four corners and record the average (compressed thickness). The strain shall be calculated as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:17Z
Check the source to verify that this is the current version before use.

$$\text{Strain (in percent)} = \frac{\text{measured thickness - compressed thickness}}{\text{measured thickness}} \times 100$$

Low tensile strength filler material shall have an average tensile strength of not less than 0.25 pounds per square inch in the weakest direction when tested in accordance with the tensile strength test below.

High tensile strength wrapping material shall have an average tensile strength of not less than 1.50 pounds per square inch in the weakest direction when tested in accordance with the tensile strength test below.

Tensile strength

Place a 4-inch square (± 1/8-inch) sample between two clamps which are 2 inches apart. The clamps shall have flat faces, a width of at least 4 inches and shall exert uniform pressure. The tensile strength shall be determined by using a universal testing machine or by suspending one clamp to a stationary support and adding weights to the other clamps until the material breaks or tears. The test shall be repeated to provide tensile strength values for the machine direction and cross direction of the material. The tensile strength shall be calculated as follows:

$$\text{Tensile strength} = \frac{\text{tensile stress (pounds)}}{\text{cross sectional area (sq.in.)}} = \frac{W}{LxT}$$

W = weight needed to break sample on tensile tester
L = length of sample (4 + 1/8-inch)
T = measured thickness

Workmanship. The cushioning material shall be clean and free from defects which might affect serviceability.

Inspection of the end item. Random samples shall be drawn from each lot for the end item inspection. Sampling shall be in accordance with MIL-STD-105.

A. Examination for construction, dimensions and workmanship. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 percent defective.

B. Testing. Use level S-2 with an AQL of 4.0 percent defective. A sample unit of 9 square feet shall be drawn from any one shipping container.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requireiments or exposing suppliers' employees to undue hazards from the recovered materials.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order. (For use by GSA; optional for other agencies) DOD may use Commercial Packaging, Standard ASTM D3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:17Z
Check the source to verify that this is the current version before use.

Note. Purchasers shall specify roll form dimensions (length, width, and thickness) and multifold units. The cushioning material may be procured in rolls 2 to 105 inches in width, 35 to 1,350 feet in length for certain thicknesses ranging from 0.25 to 2.0 inches. The purchasers shall also specify either low or high tensile strength and compressed or uncompressed material.

WHEN THIS COMMERCIAL ITEM DESCRIPTION IS USED FOR PROCUREMENT, THE COMMERCIAL ITEM CERTIFICATION CLAUSE MUST APPEAR IN THE SOLICITATION.

MILITARY INTERESTS:

Military Coordinating Activity

Army - GL

Custodians

Army - GL
Navy - AS
Air Force - 99

Review Activities

Army - AT, ME, MI
DLA - SS

PREPARING ACTIVITY:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T08:17Z
Check the source to verify that this is the current version before use.